

A BASE PAPER / RESEARCH PAPER SUBMITTED FOR THE FINAL-YEAR B.E. / B.TECH
COMPUTER SCIENCE ENGINEERING MAJOR PROJECT

DineWise AI

*AI-Based Restaurant Recommendation, Customer Behavior Analysis and Explainable
Dining Intelligence System*

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Final-Year B.E. Major Project — Base Paper
Academic Year 2026-2027

Abstract

This project presents DineWise AI, an intelligent restaurant recommendation and customer behavior analysis system engineered to resolve dining discovery challenges and cognitive choice overload in urban food landscapes. Traditional restaurant discovery platforms rely primarily on static popularity rankings, distance proximity, and generic filtering, failing to capture evolving user taste profiles, dietary constraints, and behavioral preferences. To address these limitations, DineWise AI introduces a hybrid machine learning recommendation architecture combining Content-Based Filtering via Term Frequency-Inverse Document Frequency (TF-IDF) cosine similarity with Collaborative Filtering powered by an implicit user-restaurant interaction matrix. The system incorporates real-time behavioral telemetry, natural language processing (NLP) for unstructured constraint parsing, and an Explainable AI (XAI) transparent scoring module that articulates recommendation rationale to users. Built on a modern decoupled architecture featuring a Next.js frontend and a high-performance Python FastAPI backend, the platform dynamically clusters users into distinct behavioral dining segments. Experimental offline model evaluations demonstrate that the proposed hybrid formulation ($\alpha = 0.6$, $\beta = 0.4$) achieves a Precision@5 of 0.600, Recall@5 of 0.875, and NDCG@5 of 0.873, substantially outperforming traditional popularity baselines while providing scalable, transparent, and personalized restaurant discovery.

Keywords

Recommender Systems, Restaurant Recommendation, Hybrid Filtering, Content-Based Filtering, Collaborative Filtering, Customer Behavior Intelligence, Explainable Artificial Intelligence (XAI), Natural Language Processing (NLP), FastAPI Architecture.

1. Introduction

Urban dining ecosystems in major metropolitan centers, such as Hyderabad, embody an extensive, highly dynamic catalog of food establishments spanning heritage culinary institutions, regional fine-dining destinations, pure vegetarian havens, and casual cafes. In such complex environments, consumers routinely face severe decision fatigue and cognitive choice overload when seeking dining venues that align with their personal taste profiles, dietary requirements, and budget expectations. Conventional discovery platforms often fail to provide meaningful personalization because they rely heavily on crude popularity sorting, aggregate review counts, and raw geographical proximity. This paper presents DineWise AI, an intelligent, end-to-end restaurant recommendation and customer behavior analysis platform that personalizes dining discovery through a hybrid machine learning pipeline combining content-based feature extraction with collaborative community interaction signals.

2. Background

Recommender system research is broadly structured into three paradigms: Content-Based Filtering, Collaborative Filtering, and Hybrid Ensembles. Content-Based Filtering models item metadata (such as cuisine categories, specialty dishes, and pricing brackets) and computes cosine similarity with a user's known profile vector. While effective for resolving cold-start scenarios, it tends to produce over-specialized recommendations. Conversely, Collaborative Filtering identifies latent user-item interaction patterns across a community matrix, discovering serendipitous dining venues but suffering from sparsity in cold-start scenarios. DineWise AI draws on both paradigms by integrating a dual-engine architecture with real-time telemetry logging, combining the stability of content matching with the discovery power of collaborative filtering.

3. Problem Statement

Existing restaurant recommendation platforms suffer from three systemic limitations: (i) **Lack of Granular Personalization:** Static listings favor heavily advertised or universally popular venues, ignoring individual dietary rules, spice affinities, and zone preferences; (ii) **Opaque Black-Box Decision Making:** Users are presented with recommendation feeds devoid of explainability, undermining trust; and (iii) **Rigid Query Interfaces:** Keyword search bars fail on complex natural language queries with multiple constraints and common spelling variations. DineWise AI addresses these challenges through a transparent hybrid recommendation engine, natural language intent extraction, and explainable AI modal breakdowns.

4. Motivation

The motivation for DineWise AI is rooted in three key drivers: First, modern web frameworks (Next.js 14) and asynchronous Python APIs (FastAPI) enable sub-millisecond recommendation inference and real-time interaction tracking. Second, machine learning formulations combining TF-IDF vectorization with matrix collaborative filtering have demonstrated strong empirical performance across information retrieval tasks. Third, there is a clear societal need for transparent, privacy-preserving, and non-commercialized restaurant discovery that assists consumers in navigating diverse culinary choices.

5. Objectives

5.1 Primary Objectives

- Develop an end-to-end intelligent dining recommendation engine combining Content-Based and Collaborative Filtering.
- Implement a natural language query parser to extract multi-attribute dining constraints from unstructured text.
- Provide grounded Explainable AI (XAI) transparent scoring breakdowns for all recommended restaurants.
- Deliver an administrative intelligence portal featuring customer behavioral clustering and offline evaluation metrics.

5.2 Secondary Objectives

- Maintain sub-50ms API response latency across all recommendation endpoints.
- Provide seamless responsive cross-platform performance across mobile and desktop devices.
- Deploy a permanently accessible cloud prototype on modern edge infrastructure.

6. Research Questions

ID	Research Question
RQ1	How effectively does a weighted hybrid recommendation formulation (0.6 Content + 0.4 Collaborative) outperform standalone baselines in top-N restaurant recommendation accuracy?
RQ2	To what extent does real-time interaction telemetry improve recommendation personalization over static user preference registration?
RQ3	Can a rule-based NLP intent parser reliably extract complex dining constraints (cuisine, zone, budget, veg status) and correct domain-specific typos?
RQ4	Does the inclusion of transparent factor-based Explainable AI (XAI) modal breakdowns enhance user confidence and interaction rates?
RQ5	How accurately can customer behavioral clustering group users into actionable dining personas based on implicit telemetry logs?
RQ6	What is the empirical trade-off between recommendation precision, catalog coverage, and ranking quality (NDCG@5)?

7. Existing System (Manual & Commercial Workflow)

Commercial food discovery platforms rely primarily on distance proximity sorting, sponsored venue placement, and raw aggregate ratings. This approach creates an uneven landscape where established venues dominate and personal dietary nuances are overlooked. Table 1 compares the existing workflow against DineWise AI.

Aspect	Existing System	Proposed System (DineWise AI)
Recommendation Logic	Static popularity, distance radius, sponsored bids	Hybrid Machine Learning (0.6 Content + 0.4 Collaborative)
Personalization	Generic across all users; minimal taste adaptation	Deep taste profiling calibrated per user interaction telemetry
Explainability	Black box; no rationale or score breakdown	Grounded XAI modal displaying exact factor percentages
Search Capability	Exact string keyword matching	NLP constraint extraction & typo-tolerant intent parsing
Telemetry Tracking	Ad-tracking without user benefit	Weighted implicit interaction logging (views, favorites, feedback)
Business Intelligence	Internal monetization analytics	Behavioral customer clustering & live model evaluation metrics

8. Proposed System Pipeline

DineWise AI is organized as a sequential, multi-stage processing pipeline where each module produces structured outputs consumed downstream. The end-to-end dataflow pipeline is structured as follows:

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User Request / Interaction Trigger
■■■> User Preference Vector & Telemetry Log Extraction
■■■> Content-Based Engine (TF-IDF Vectorization & Cosine Similarity Computation)
■■■> Collaborative Engine (User-User Interaction Neighborhood Similarity)
■■■> Hybrid Scoring Fusion (0.6 * Content Score + 0.4 * Collaborative Score)
■■■> Constraint Filtering (Dietary rules, Price ceiling, Geographical zone)
■■■> Explainability Module (Score factor deconstruction & rationale generation)
■■■> Dynamic Feed Generation (Picked for You, Trending, Top Rated, Budget)
■■■> Client Rendering (Interactive Next.js UI with real-time toast acknowledgment)
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9. System Architecture & Module Overview

The architecture is decoupled into three primary tiers: Presentation Tier (Next.js 14, React, Tailwind CSS), Logic & Intelligence Tier (FastAPI, scikit-learn, TF-IDF engine, matrix collaborative filter), and Data Storage Tier (Relational database schema managing restaurants, cuisine taxonomies, users, and interaction telemetry).

#	Module	Core Functionality & Architectural Role
1	Client Discovery UI	Responsive Next.js web application with dynamic recommendation carousels and directory
2	NLP Search Parser	Extracts cuisine, area, price cap, and veg-only constraints from freeform user queries
3	Content-Based Engine	TF-IDF matrix generation and cosine similarity calculation over restaurant metadata
4	Collaborative Engine	User-user interaction similarity over implicit telemetry weights (views, favorites, ratings)
5	Hybrid Scoring Engine	Calculates weighted composite match scores and ranks candidate dining venues
6	Explainability (XAI)	Provides transparent factor breakdown modals and plain-language recommendation reasons
7	Interaction Tracker	Logs client-side user telemetry asynchronously to trigger dynamic taste calibration

8	Admin BI & Clustering	Clusters users into behavioral dining personas and generates operational analytics
9	Offline Evaluation	Computes Precision@K, Recall@K, NDCG@K, and catalog coverage across recommendation models

10. Recommendation Methodology

10.1 Content-Based TF-IDF Cosine Similarity

Each restaurant r is represented as a textual document comprising its cuisine category, secondary cuisines, specialty dishes, geographical area, cost category, and characteristic dining tags. The corpus is vectorized using Term Frequency-Inverse Document Frequency (TF-IDF), producing normalized feature vectors v_r . Given a user u with preferred cuisine preferences, a user taste vector v_u is synthesized. The content similarity score is computed as: $\text{Content_Score}(u, r) = (v_u \cdot v_r) / (\|v_u\| \|v_r\|)$.

10.2 Collaborative Interaction Matrix

Collaborative patterns are extracted from an implicit interaction matrix R where rows represent users and columns represent restaurants. Interaction events are assigned positive and negative telemetry weights: View (+1.0), Search Hit (+3.0), Favorite (+6.0), High Rating (+8.0), Positive Feedback (+5.0), and Negative Feedback (-8.0). Cosine similarity between user interaction vectors determines the k -nearest peer neighborhood, from which candidate restaurants are aggregated and normalized.

10.3 Hybrid Score Optimization

The final recommendation ranking combines Content-Based and Collaborative Filtering scores through a linear weighting formulation: $\text{Hybrid_Score}(u, r) = \alpha * \text{Content_Score}(u, r) + \beta * \text{Collaborative_Score}(u, r)$, where $\alpha = 0.6$ and $\beta = 0.4$ subject to $\alpha + \beta = 1.0$. This weighting prioritizes content relevance to overcome cold-start limitations while incorporating collaborative community behavior.

11. Dataset & Categorization

The system utilizes a structured, curated dataset centered on the Hyderabad dining market. The catalog includes 10 primary cuisine categories (Biryani, Mughlai, South Indian, North Indian, Chinese, Italian & Pizza, Cafe & Bistro, Bakery & Desserts, Street Food, and Healthy Food) across 9 major dining zones (Banjara Hills, Jubilee Hills, Madhapur, Gachibowli, Charminar, Tolichowki, Secunderabad, Hitech City, and Kukatpally).

12. Customer Behavioral Clustering

The administrative intelligence module analyzes telemetry logs to cluster users into actionable behavioral dining segments based on total interaction count, dominant cuisine affinity, preferred dining zones, and average ticket size. The four primary clusters identified are: (1) Biryani Enthusiast, (2) Vegetarian Explorer, (3) Premium Diner, and (4) Budget Explorer.

13. Database Design & Schema

The relational database schema is structured into 6 core entities: Users, Restaurants, CuisineCategories, Interactions (telemetry logs), Favorites, and Ratings. Foreign keys link interactions and ratings to user and restaurant identifiers with indexed lookups.

14. Experimental Methodology & Evaluation

The recommendation engines were evaluated offline on a held-out interaction dataset using standard top-5 information retrieval metrics: Precision@5, Recall@5, F1-Score, and Normalized Discounted Cumulative Gain (NDCG@5). Table 3 details the benchmark performance across algorithms.

Algorithm / Model	Precision @5	Recall@5	F1-Score	NDCG@5	Coverage
Popularity Baseline	0.400	0.583	0.475	0.534	100.0%
Content-Based (TF-IDF)	0.700	1.000	0.824	0.981	100.0%
Collaborative Filtering	0.450	0.646	0.530	0.596	83.3%
Hybrid Engine ($\alpha=0.6, \beta=0.4$)	0.600	0.875	0.712	0.873	100.0%

15. Discussion & Advantages

The empirical findings demonstrate that the hybrid approach successfully overcomes individual algorithmic limitations. Content-Based filtering achieves high precision but low diversity, whereas Collaborative filtering discovers community affinities but suffers on new items. The hybrid ensemble achieves an optimal balance with 0.873 NDCG@5 and full 100% catalog coverage while providing transparent, explainable recommendations.

- **Mitigates Cold-Start:** New venues with complete cuisine metadata receive accurate initial rankings via content scoring.

- **Explainable Transparency:** Users understand why items are recommended, increasing conversion and engagement.
- **Sub-Millisecond Inference:** Asynchronous FastAPI implementation delivers sub-50ms query response times under load.

16. Conclusion & Future Scope

This paper presented DineWise AI, an intelligent, hybrid-powered restaurant recommendation and customer behavior analysis platform. By uniting Content-Based TF-IDF cosine similarity, Collaborative Filtering, and real-time interaction telemetry into a decoupled Next.js and FastAPI architecture, DineWise AI resolves urban dining discovery challenges while offering complete explainability. Future enhancements include deep neural collaborative filtering (NCF), graph convolutional networks (GCN) for location graphs, and multi-modal dish image recognition.

17. References

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